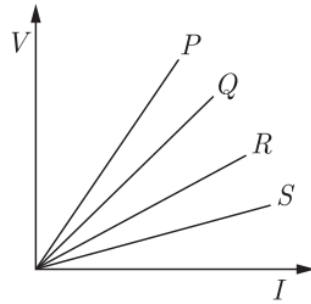


Chapter

Electricity and Magnetism

1. The graph of voltage vs current for four different materials is shown below.

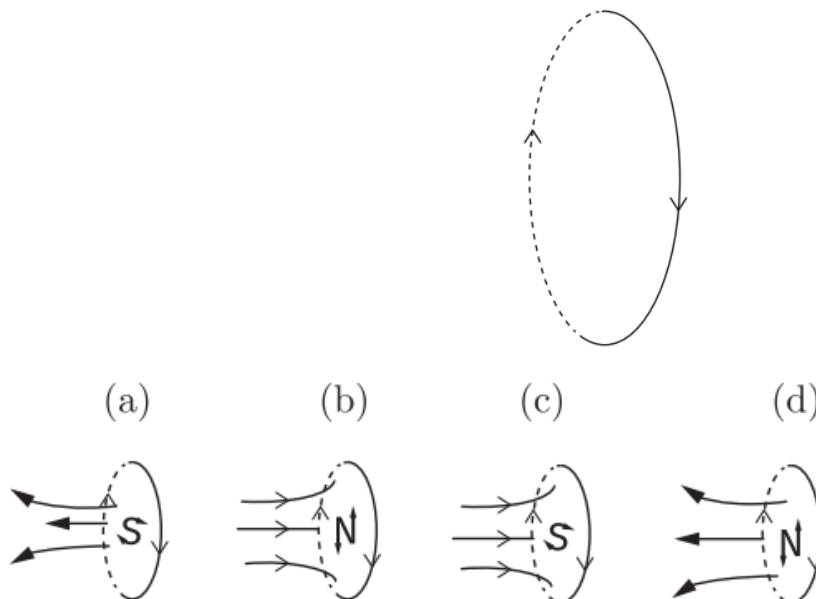


Which of these four materials would be used for making filament of a bulb?

- (a) Q (b) S
(c) P (d) R

SQP2025

2. Current is flowing through a coil as shown in the figure. Which one of the given figures will correctly depict the magnetic polarity and the direction of the lines of force along the axis of the coil.



SQP2025

3. According to the old convention, the colour of the earth wire is:

- (a) black
- (b) green
- (c) yellow
- (d) red

SQP2025

4. Two copper wires can have the different [resistivity / resistance] but will have same [resistance / resistivity]

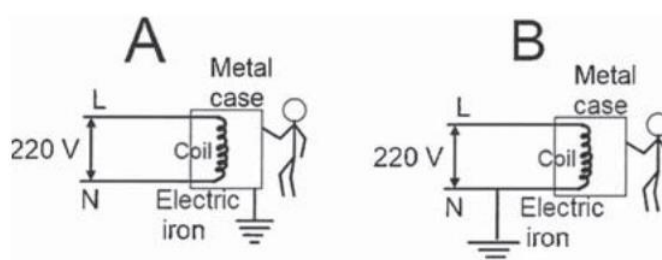
SQP2025

5. (a) Name the principle of AC generator.

(b) State its one use.

SQP2025

6. If live wire makes an accidental contact with the metal case, which circuit (A or B) in the diagram, illustrating an electric iron, is considered safe for the user (Assuming the fuse is present in the live wire in both circuits)? Justify your answer.



SQP2025

7. A transformer is used to change a high alternating e.m.f. to a low alternating e.m.f. of the same frequency.

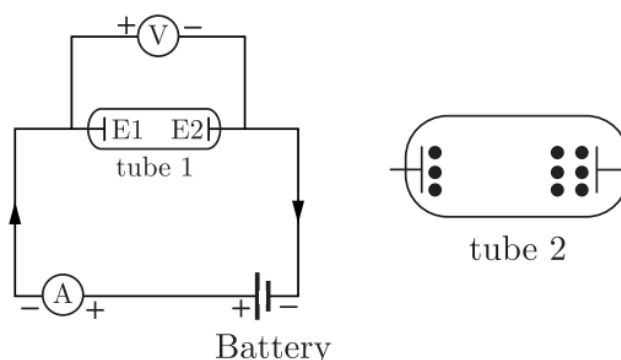
(a) Identify the type of transformer used for the above purpose.

(b) State whether the turns ratio of the above transformer is $= 1$ or > 1 or < 1 .

SQP2025

8. The circuit depicted in the figure is employed for studying Ohm's Law. Instead of using a standard resistor, a student opts for a glass tube filled with mercury (tube 1), connected to the circuit through two electrodes E1 & E2. He records the readings of the ammeter and voltmeter, thereby calculates the resistance. The student repeats the experiment by substituting tube 1 with

tube 2, where the same amount of mercury fills the tube 2.

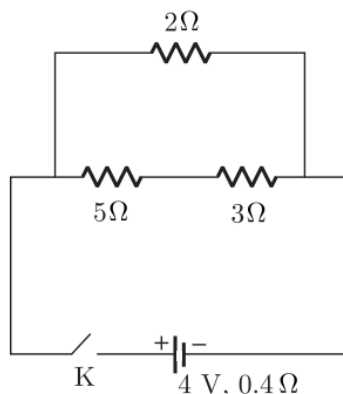


Neglecting internal resistance of the cell use (> or < or =) to compare

- (a) the resistance in both the cases.
- (b) the voltmeter readings in both the cases.
- (c) the specific resistance in both the cases.

SQP2025

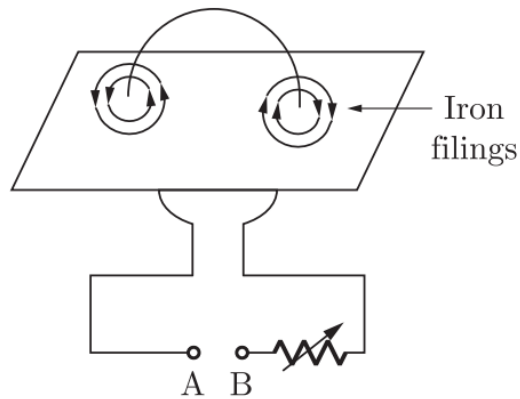
9. Observe the given circuit diagram and answer the questions that follow :



- (a) Calculate the resistance of the circuit when the key K completes the circuit.
- (b) Calculate the current through 3Ω resistance when the circuit is complete.

SQP2025

10. The diagram below shows a cardboard on which iron filings are kept. A wire bent in the form of a loop is seen passing through the cardboard. When current flows through it the iron filings arrange themselves as shown below with the direction of magnetic field.



(a) State the polarities of the battery at A and B.

(b) State the effect on the magnetic field if an iron rod is held along the axis of the coil.

(c) State one way to:

(1) change the polarity of the coil.

(2) decrease the strength of the magnetic field around the coil.

SQP2025

11. The specific resistance of a conductor depends on its

- (a) length
- (b) material
- (c) area of cross section
- (d) radius

MAIN 2024

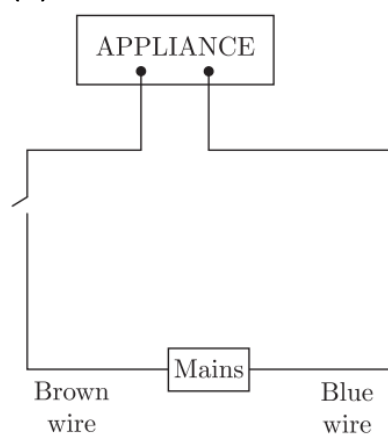
12. The potential difference between terminals of a cell in a closed electric circuit is

- (a) terminal voltage
- (b) electro motive force
- (c) voltage drop
- (d) none of these

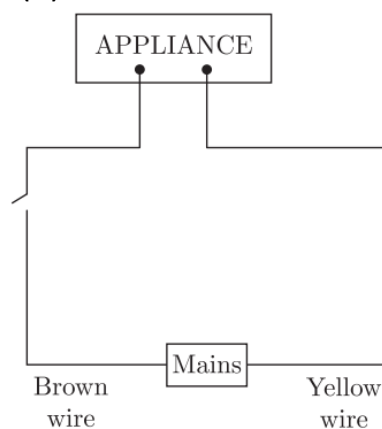
MAIN 2024

13. Identify the option that displays the correct wiring with correct colour code

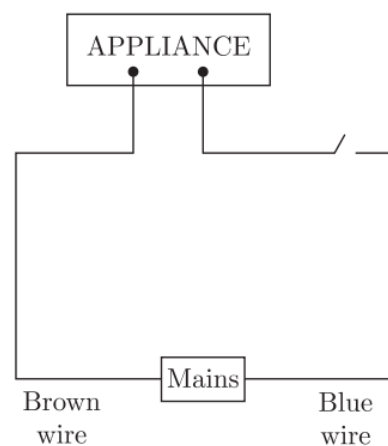
(a)



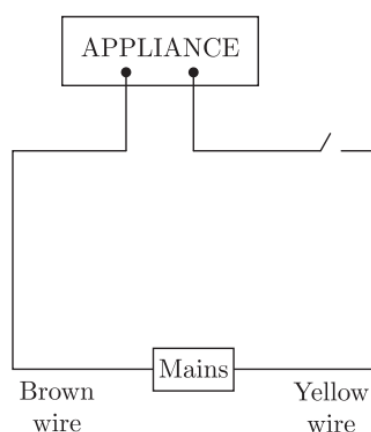
(b)



(c)



(d)



MAIN 2024

14. A fuse is rated 5 A. Can it be used with a geyser rated 1540 W, 220 V. Write Yes or No. Give supporting calculations to justify your answer.

MAIN 2024

15. 5 bulbs are connected in series in a room. One bulb is fused. It is removed and remaining 4 bulbs are again connected in series to the same circuit. What will be the effect on the following physical quantities? (Increases, Decreases, Remain Same)

(a) Resistance

(b) Intensity of light

MAIN 2024

16. State two factors affecting the speed of rotation of the coil in a D.C. motor.

MAIN 2024

17. The voltage - current readings of a certain material are shown in the table given below:

Voltage (V)	10 V	20 V	30 V
Current (I)	2 A	3 A	4 A

Study the table.

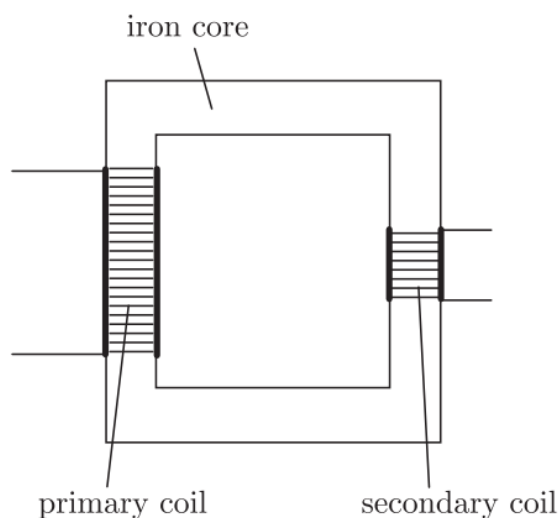
(a) State whether the conductor used is ohmic or non-ohmic.

(b) Justify your answer.

(c) State Ohm's law.

MAIN 2024

18. Below is the diagram of a transformer:

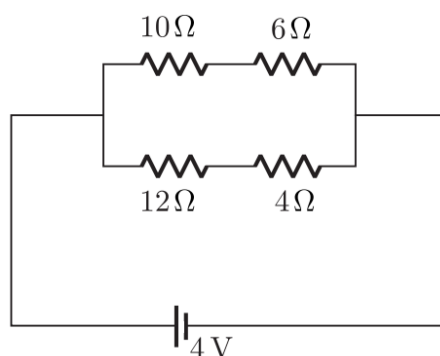


(a) Identify the type of transformer.

(b) In this type of transformer which of the wire is thicker, the primary or the secondary? Give a reason.

MAIN 2024

19. Study the diagram:



(a) Calculate the total resistance of the circuit.

(b) Calculate the current drawn from the cell.

(c) State whether the current through 10 Ω resistor is greater than, less than or equal to the current through the 12 Ω resistor.

MAIN 2024

20. Bends in a rubber pipe reduce the flow of water through it. How would the bends in a wire affect its electric resistance ?

SQP 2023

21. An immersion heating rod is rated 220 V and can bring certain amount of water to its boiling point in 15 min. When this immersion rod is actually connected to an electric circuit, it brings the water to boil in 18.15 min. Calculate the line voltage.

(a) 805 V

(b) 300 V

(c) 560 V

(d) 200 V

MAIN 2023

22. What purpose is served by the terminals of a three way in plug?

MAIN 2023

23. (a) What will happen to a compass needle when the compass is placed below a wire and a current is made to flow through the wire?

Give a reason to justify your answer.

(b) What energy conversion takes place during the working of a D.C. motor?

MAIN 2023

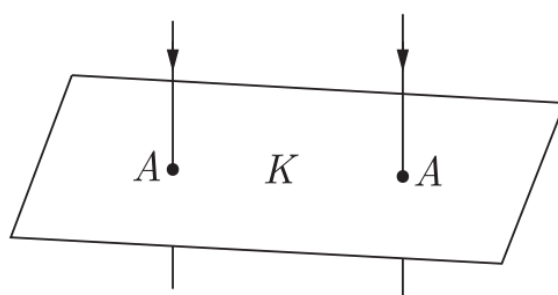
24. You are provided a cell, an insulated copper coil, the soft iron bar and a switch. Draw a circuit diagram to represent the process of making an electromagnet.

MAIN 2023

25. Distinguish between terminal potential difference and emf of a cell.

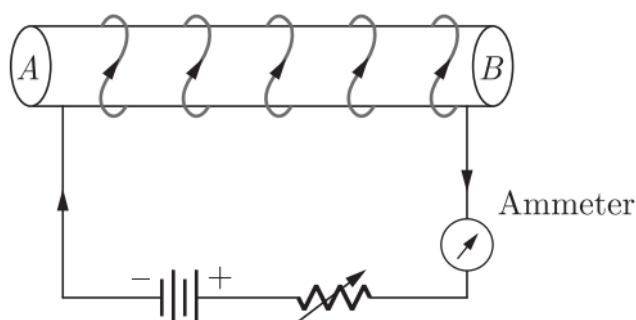
COMP 2023

26. The following diagram shows two straight wires carrying current. Copy the diagram and draw the pattern of lines of force around them and mark their directions.



COMP 2023

27. Diagram below shows a circuit containing a coil wound over a long and thin hollow cardboard tube. Copy the diagram. (i) Show the polarity acquired by each face of the solenoid. (ii) Draw the magnetic field lines of force inside the coil and also show their direction. (iii) Mention two methods to increase the strength of the magnetic field inside the coil.

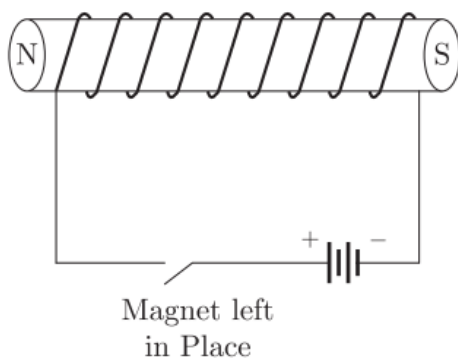


COMP 2023

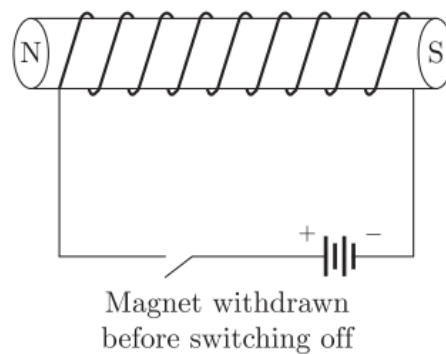
28. A permanent magnet can be demagnetised by using a solenoid and switching the current on then off.

Which diagram shows the most effective method of producing demagnetisation ?

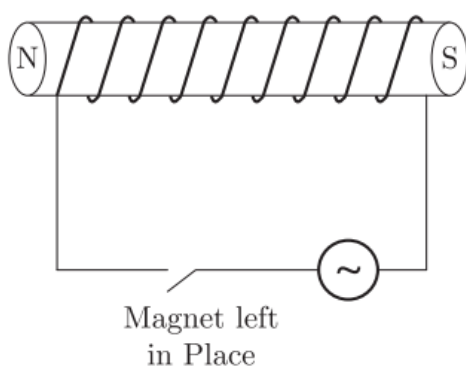
(a)



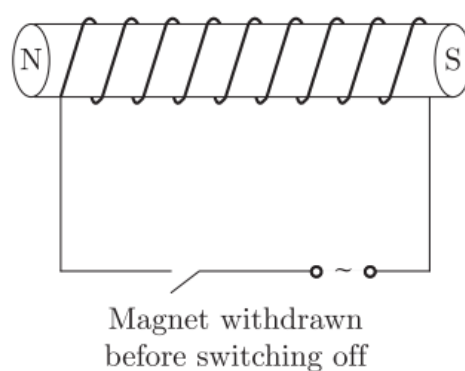
(b)



(c)



(d)



SQP 2022

29. What special characteristics must :

(a) A heating wire and

(b) Fuse wire have?

SQP 2022

30. What is an electromagnet? What do you know about the simplest form of an electromagnet?

SQP 2022

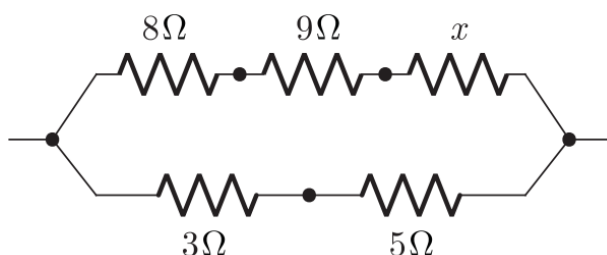
31. What will happen if switch is connected in 'neutral wire'?

MAIN 2022

**32. What is the commercial unit of electrical energy? Convert it into joule.
What is the other name of the commercial unit?**

MAIN 2022

33. Equivalent resistance of circuit diagram is $6\ \Omega$. Calculate the value of x .



(a) $8\ \Omega$

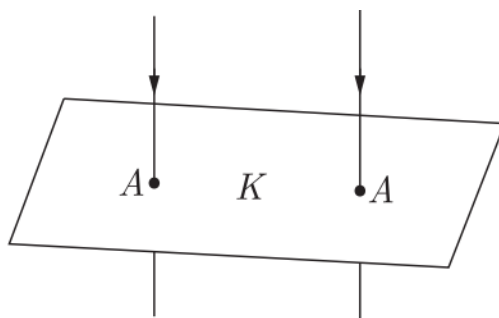
(b) $7\ \Omega$

(c) $9\ \Omega$

(d) $10\ \Omega$

MAIN 2022

34. The following diagram shows two straight wires carrying current.



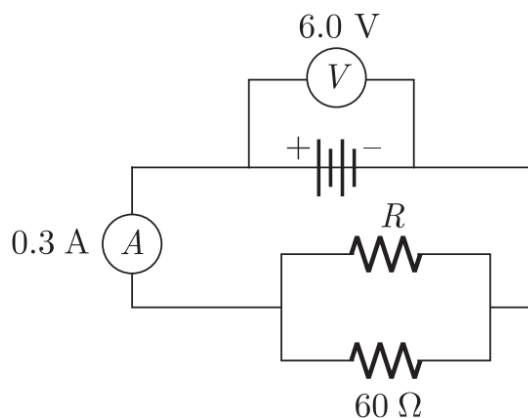
Copy the diagram and draw the pattern of lines of force around them and mark their directions.

MAIN 2022

35. Explain briefly how a magnet can be demagnetised using an alternating current.

MAIN 2022

36. In given the figure below, the ammeter A reads 0.3 A .



Calculate:

- (a) the total resistance of the circuit.**
- (b) the value of R .**
- (c) the current flowing through R .**

MAIN 2022

37. The graph plotted for potential difference (V) against current (I) for ohmic resistors is

- (a) a curve passing through the origin
- (b) a straight line not passing through origin
- (c) a straight line passing through origin
- (d) a circle centred at the origin

SEM-II 2022

38. A main switch in the main distribution board is present in

- (a) A live wire
- (b) a neutral wire
- (c) a live as well as neutral wire
- (d) an earth wire

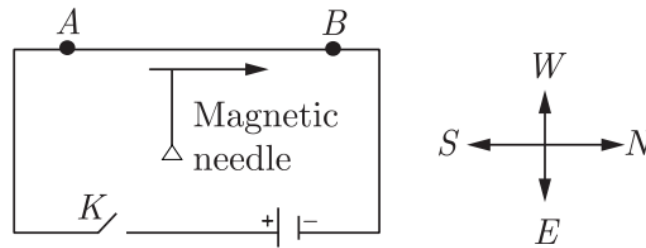
SEM-II 2022

39. A coil wound around a piece of soft iron can become an electromagnet only when

- (a) the circuit is open
- (b) a magnetic compass is present in the vicinity
- (c) a galvanometer is connected to the circuit
- (d) a current flows in the circuit

SEM-II 2022

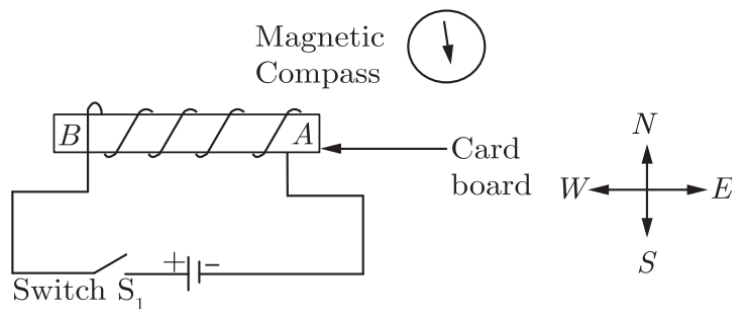
40. A conductor AB is kept along North-South direction of the earth above a magnetic needle as shown below. When the key K is closed, then



- (a) the needle will not show any deflection.
- (b) the needle will deflect towards East.
- (c) the needle will turn in the opposite direction i.e. towards South.
- (d) the needle will deflect towards West.

SEM-II 2022

41. The diagram below shows a magnetic compass kept closer to a coil AB wound around a hollow cylindrical cardboard.



- (i) After studying the circuit and the magnetic compass carefully, state whether the switch S₁ is open or closed.
- (ii) How did you arrive at the conclusion in (i)?
- (iii) What is the purpose of placing the magnetic compass in the above setup?

SEM-II 2022

42. If a wire of resistance 2 Ω gets stretched to thrice its original length

- (i) Calculate the new resistance of the wire.
- (ii) What happens to the specific resistance of the wire?

SEM-II 2022

43. What are the factors on which heat dissipated by a conductor depends?

COMP 2022

44. Explain:

- (i) Why are the pole pieces made cylindrical?**
- (ii) Why is a soft iron core used within the coil?**

COMP 2022

45. What is a solenoid? Explain with the help of a figure how a solenoid behaves like a bar magnet?

COMP 2022

46. A current of 4 A flows through a 12 V car head light bulb for 10 minutes. How much energy transfer takes place during this time?

COMP 2022

47. What is meant by MCB ? Which part of the domestic electric circuit does it replace?

SQP 2021

48. How does the resistivity of a

- (i) metallic wire,**
- (ii) semiconductor,**
- (iii) an alloy such as constantan depend on temperature?**

SQP 2021

49. Two resistors of 3 ohm and 6 ohm are connected in parallel. What is the equivalent resistance ?

SQP 2021

50. A bulb is joined to a battery of e.m.f. 6 V. A steady current of 0.5 A flows through the circuit. Calculate the total energy provided by the battery in 5 minutes.

SQP 2021

51. Draw a schematic diagram of a circuit consisting of a battery of four 2 volt cells, a 5 ohm resistor, an 8 ohm resistor and a 12 ohm resistor and a plug key all connected in series.

SQP 2021

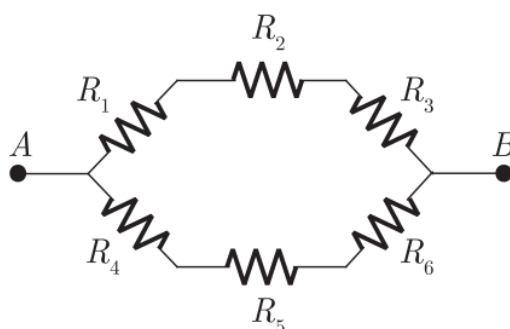
52. State Ohm's law.

MAIN 2021

53. Diagrammatically illustrate how you would connect a key, a battery, a voltmeter, an ammeter, an unknown resistance R and a rheostat so that it can be used to verify the above law.

MAIN 2021

54. Calculate the equivalent resistance of the circuit as shown in figure, in which six equal resistances of 1 ohm each are connected in hexagonal form.



MAIN 2021

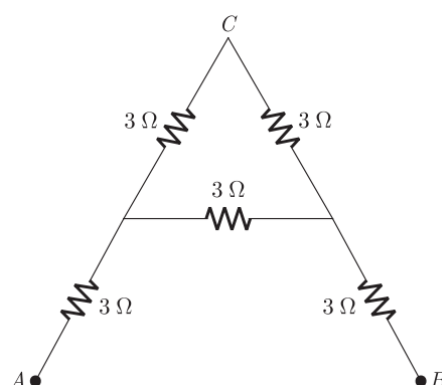
55. A house is fitted with a motor of one horse power, 5 bulbs consuming 0.2 A each and two fans of 80 W each. If all the appliances work for five hours daily, calculate the monthly bill. Energy costs 20 paise per unit. The energy is supplied at 220 V. Given 1 hp = 746 W.

MAIN 2021

56. Obtain the expression for the electrical energy produced in a circuit.

MAIN 2021

57. In the diagram given below, calculate the resistance between the points (i) X and Y, (ii) A and B.



MAIN 2021

58. The power of a d.c. motor can be increased :

- (a) by laminating its soft iron core
- (b) by increasing the strength of current flowing through it
- (c) by increasing number of turns in its coil
- (d) all of these

COMP 2021

59. In an electric motor :

- (a) electric energy changes to mechanical energy.
- (b) mechanical energy changes to heat energy.
- (c) mechanical energy changes to electric energy.
- (d) electric energy changes to magnetic energy

COMP 2021

60. When is the potential difference between two points said to be 1 volt?

COMP 2021

61. Define electrostatic potential at a point.

COMP 2021

62. How can an electric heater designed for 220 V be adopted for 110 V without changing the length of the coil and also without a change in the consumed power?

COMP 2021

63. Write four differences between electromotive force and potential difference.

COMP 2021

64. What do you mean by the following terms :

- (a) Potential ?
- (b) Electromotive force ?
- (c) Potential difference ?

COMP 2021

65. Distinguish between the resistors connected in series and in parallel.
Giving circuit diagram.

COMP 2021

66. Four lamps of 200 watt, 100 watt, 60 watt and 40 watt are connected to a power supply of 220 volts; calculate :

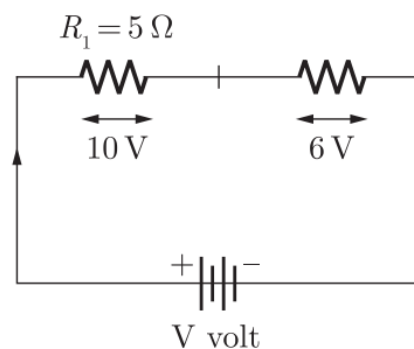
(i) total current consumed.

(ii) total resistance of this arrangement and

(iii) the cost of keeping them lighted for 7 hours daily for 30 days, the cost of electricity being 30 paise per unit.

COMP 2021

67. Two resistors are connected in series as shown in the diagram :



(i) What is the current through the 5 ohm resistor ?

(ii) What is the current through R ?

(iii) What is the value of R ?

(iv) What is the value of V ?

COMP 2021

68. Redraw the circuit of the previous question, putting in an ammeter to measure the current through the resistors and a voltmeter to measure the voltage across the 12 ohm resistor. What is the reading of the ammeter and the voltmeter ?

COMP 2021

69. A heating coil is immersed in a calorimeter of heat capacity $50 \text{ J } ^\circ \text{C}^{-1}$ containing 1.0 kg of a liquid of specific heat capacity $450 \text{ J kg}^{-1} ^\circ \text{C}^{-1}$. The temperature of liquid rises by 10°C when 2.0 A current is passed for 10 minutes. Find: (i) the resistance of the coil, (ii) the potential difference across the coil. State the assumption used in your calculations.

COMP 2021

Solution

1.

Ans: (c) P

Explanation:

A bulb filament must have **high resistance** so that it produces a strong heating effect and glows.

In the V–I graph, the **slope of the line represents resistance** ($R = V/I$). Material **P** has the **steepest slope**, meaning it has the **highest resistance**.

Therefore, **P** is the best material for making the filament of a bulb.

2.

Ans: (c)

Explanation:

Use the **clock-face rule**:

- Look at the face of the coil from one side.
- If the current in that face is **anticlockwise**, that face behaves as a **North pole**.
- If the current is **clockwise**, that face behaves as a **South pole**.

From the given direction of current in the coil, one side of the coil is North and the opposite side is South, as shown in option (c).

Magnetic field lines always go **out of the North pole** and **into the South pole** along the axis, which also matches diagram (c).

3.

Ans: (b) green

Explanation:

In the old wiring colour convention:

- The **live wire** was **red**.
- The **neutral wire** was **black**.

- The **earth wire** was **green**.

Therefore, the earth wire's colour according to the old convention is **green**.

4.

Ans: resistance, resistivity

5.

Ans: (a) The AC generator works on **Faraday's law of electromagnetic induction**.

(b) Use: It is used as a **backup power source** in buildings during electricity failure. (Any correct use is acceptable.)

6.

Ans: Circuit A.

In **circuit A**, the metal case is **earthed**, so if the live wire touches the metal case, the current flows to the earth and **the user does not get an electric shock**.

In **circuit B**, the metal case is **not earthed**, so the current can pass through the person's body, giving an **electric shock**.

7.

Ans: (a) Step-down transformer.

(b) The turns ratio is **less than 1**.

8.

Ans: (a) Resistance of tube 2 < resistance of tube 1.

(b) The voltmeter reading for tube 1 is **equal to** the voltmeter reading for tube 2.

(c) The specific resistance (resistivity) is the **same** in both cases.

9.

Ans: (a) Resistance of 5 Ω and 3 Ω resistors in series:

$$R_1 = 5 + 3 = 8 \Omega$$

This 8 Ω is in **parallel** with 2 Ω :

$$R_2 = \frac{8 \times 2}{8 + 2} = \frac{16}{10} = 1.6 \, \Omega$$

Including internal resistance $0.4 \, \Omega$:

$$R = 1.6 + 0.4 = 2 \, \Omega$$

(b) Total current in the circuit:

$$I = \frac{E}{R} = \frac{4}{2} = 2 \, \text{A}$$

Current through the $3 \, \Omega$ resistor (in the $8 \, \Omega$ branch):

$$I_{3\Omega} = \frac{2 \times 2}{10} = 0.4 \, \text{A}$$

10.

Ans:

(a) Polarity of the battery:

- **A = Positive terminal**
- **B = Negative terminal**

(b) When an iron rod is held along the axis of the coil, the **magnetic field becomes stronger** because the iron rod increases the magnetic flux through the coil.

(c)

(1) To change the polarity of the coil:

- **Interchange the terminals A and B** (i.e., reverse the direction of current in the coil).

(2) To decrease the strength of the magnetic field:

- **Decrease the current** flowing through the coil.

11.

Ans: (b) material

Explanation:

Specific resistance (or resistivity) is an **intrinsic property** of a material.

It depends **only on the nature of the material** and **does not change** with the length, radius, or area of the wire.

So, specific resistance depends only on the **material** of the conductor.

12.

Ans: (a) terminal voltage

Explanation:

The potential difference across the terminals of a cell:

- **In open circuit** → called **EMF (electromotive force)**
- **In closed circuit (current flowing)** → called **terminal voltage**

So, in a closed circuit, the potential difference is the **terminal voltage**.

13.

Ans: (a)

Explanation:

According to the correct colour code:

- **Blue wire** → Neutral (N)
- **Brown wire** → Live (L)

In proper wiring:

- The **neutral (blue) wire** should go **directly** to the appliance.
- The **live (brown) wire** should pass **through a switch** before reaching the appliance for safety.

Option **(a)** shows this correct arrangement.

14.

Ans: The current drawn by the geyser is

$$I = \frac{P}{V} = \frac{1540}{220} = 7 \text{ A}$$

Since the geyser needs **7 A**, a **5 A fuse** will blow.

Therefore, the 5 A fuse cannot be used.

15.

Ans:

(a) Resistance – Decreases

When one bulb is removed, the number of bulbs in series reduces from 5 to 4.

Since total resistance in series is the sum of individual resistances, the total resistance **decreases**.

(b) Intensity of light – Increases

With lower total resistance, a larger current flows through the circuit.

Hence, the remaining bulbs glow brighter and the **intensity of light increases**.

16.

Ans: Two factors that affect the speed of rotation of the coil in a D.C. motor are:

1. **Increasing the number of turns of the armature coil** increases the speed of rotation.
2. **Increasing the applied battery e.m.f. (voltage)** also increases the speed of rotation.

17.

Ans: (a) The conductor is non-ohmic.

(b) The resistance of the material is not constant; it increases as the voltage and current increase.

(c) Ohm's law states that the current flowing through a conductor is directly proportional to the potential difference across it, provided the physical conditions of the conductor (such as temperature, pressure, density, volume, etc.) remain constant.

18.

Ans: (a) It is a **step-down transformer**.

(b) The **secondary coil has thicker wire** because the **secondary current is greater than the primary current**, and a thicker wire is needed to safely carry the higher current.

19.

Ans:

(a) Total resistance of the circuit

The 10 Ω and 6 Ω resistors are in series:

$$R_1 = 10\Omega + 6\Omega = 16\Omega$$

Similarly, the 12 Ω and 4 Ω resistors are in series:

$$R_2 = 12\Omega + 4\Omega = 16\Omega$$

These two combinations R_1 and R_2 are connected in parallel.

Hence, the equivalent resistance is:

$$R_{eq} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{16 \times 16}{16 + 16} = \frac{256}{32} = 8\Omega$$

(b) Current drawn from the cell

$$I = \frac{V}{R_{eq}} = \frac{4\text{ V}}{8\Omega} = 0.5\text{ A}$$

(c) Comparison of current through 10 Ω and 12 Ω resistors

Each parallel branch has the **same total resistance** (16 Ω).

Therefore, the current in both branches is equal.

So, the current through the 10 Ω resistor is **equal to** the current through the 12 Ω resistor.

20.

Ans: Bends in the wire do **not** affect its resistance. Electrons are extremely small compared to the bends, so they can easily change their direction of motion without any obstruction.

21.

Ans: (d) 200 V

Explanation:

For the same mass of water brought to boiling, the heat (energy) supplied is the same in both cases.

Heat supplied $\propto$ Power $\times$ Time, and

$$P = \frac{V^2}{R}$$

So,

$$\frac{220^2}{R} \times 15 = \frac{V^2}{R} \times 18.15$$

The resistance R cancels out:

$$220^2 \times 15 = V^2 \times 18.15 \Rightarrow V^2 = 40000 \Rightarrow V = 200 \text{ V}$$

Hence, the line voltage is **200 V**.

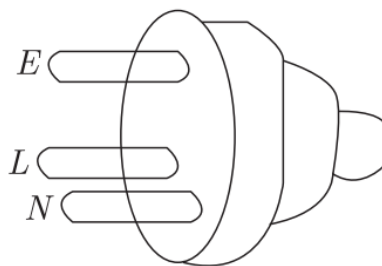
22.

Ans: In a three-pin plug, the three terminals serve different purposes:

- **L** is for the *live wire*,
- **N** is for the *neutral wire*, and
- **E** is for *earthing*.

The earth pin is made thicker so that it cannot be inserted into the live or neutral slots.

It is also made longer so that it connects to the earth terminal first, ensuring safety and preventing electric shock to the user.

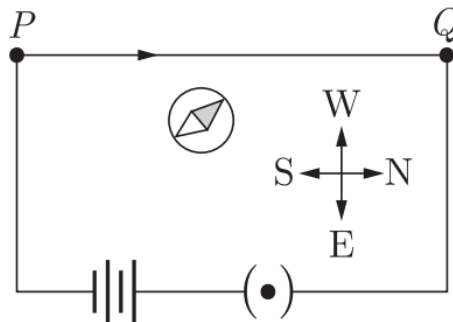


23.

Ans:

(a) According to Oersted's experiment, when a compass is placed below a current-carrying wire, the magnetic field produced by the current causes the compass needle to deflect.

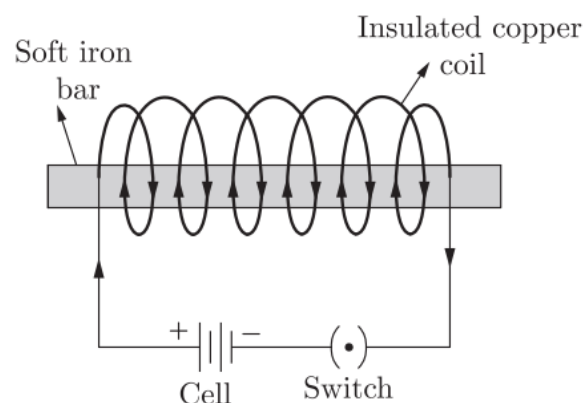
If the current flows from south to north, the **north pole of the needle will deflect towards the west**. This happens because a current-carrying conductor produces a magnetic field around it.



(b) During the working of a D.C. motor, **electrical energy is converted into mechanical energy**.

24.

Ans:



25.

Ans:

(i) Terminal potential difference:

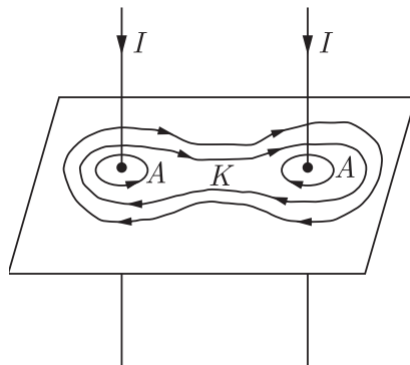
It is the potential of a cell **when current is being drawn from it**. In other words, it is the potential of the cell **in a closed circuit**.

(ii) EMF:

It is the potential of a cell **when no current is being drawn**. In other words, it is the potential of the cell **in an open circuit**.

26.

Ans: The pattern of lines is as shown in the figure below :

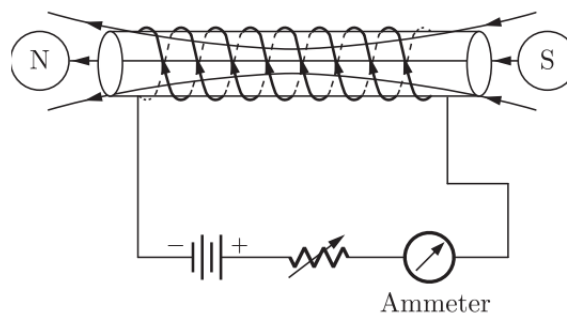


27.

Ans:

(i) The polarity acquired by the two ends is as shown below. (A shows North and B shows South polarity)

(ii)

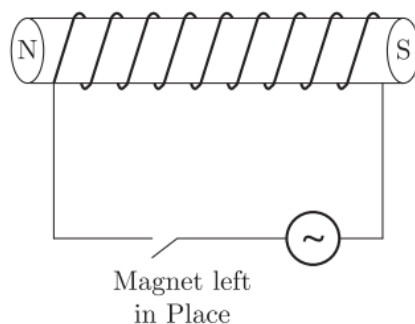


(iii) Increase the strength of current, increase the number of turns in the coil, insert soft iron rod in the coil.

28.

Ans: (c)

Explanation:



To demagnetise a permanent magnet using a solenoid, we pass **alternating current (AC)** through the solenoid and **slowly reduce it to zero while the magnet is still inside**.

AC keeps reversing direction, so the magnetic domains in the magnet keep flipping and finally become randomly arranged. When the current becomes zero, the magnet is left with almost no net magnetisation.

If the magnet is pulled out *before* switching off the current (as in some other options), it comes out while still strongly magnetised.

So, the most effective method is shown in **diagram (c)**.

29.

Ans: (a) A heating wire should have **high resistivity** and a **high melting point**.

(b) A fuse wire should have **high resistance** and a **low melting point** so that it melts quickly when excess current flows.

30.

Ans: An electromagnet is a **temporary magnet** that can be magnetised or demagnetised very quickly as needed.

In its simplest form, an electromagnet consists of a **soft iron core** around which a **large number of turns of insulated wire** are wound.

31.

Ans: If the switch is connected in the **neutral wire**, the appliance (for example, an electric iron) will still remain connected to the **live wire** even when the switch is OFF.

Therefore, if someone touches the appliance, he or she may receive an electric shock, which can sometimes be fatal.

32.

Ans: The commercial unit of electrical energy is the **kilowatt-hour (kWh)**.

$$1 \text{ kWh} = 1 \text{ kW} \times 1 \text{ hour}$$

$$= 1000 \text{ W} \times 3600 \text{ s}$$

$$1000 \text{ J s}^{-1} \times 3600 \text{ s}$$

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

The other name of this commercial unit is the **Board of Trade Unit (BOTU)**.

33.

Ans: The resistors $8\ \Omega$, $9\ \Omega$, and $x\ \Omega$ are in series:

$$R_1 = 8 + 9 + x = (17 + x)\ \Omega$$

The resistors $3\ \Omega$ and $5\ \Omega$ are also in series:

$$R_2 = 3 + 5 = 8\ \Omega$$

These two combinations R_1 and R_2 are connected in parallel, and their equivalent resistance is given as $6\ \Omega$.

$$\frac{1}{6} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{6} = \frac{1}{17 + x} + \frac{1}{8}$$

$$\frac{1}{17 + x} = \frac{1}{6} - \frac{1}{8}$$

$$\frac{1}{17 + x} = \frac{4 - 3}{24} = \frac{1}{24}$$

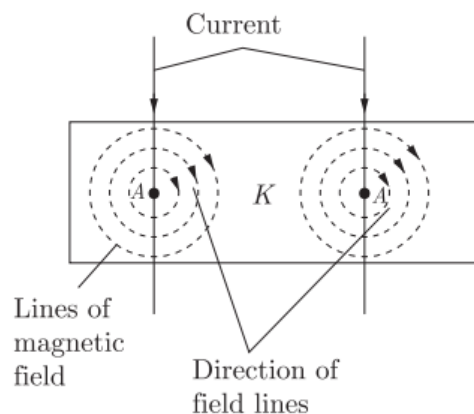
$$17 + x = 24$$

$$x = 24 - 17 = 7\ \Omega$$

Thus, the correct answer is **$7\ \Omega$** .

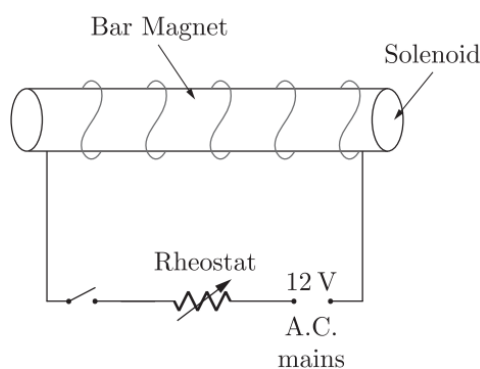
34.

Ans: The direction of field is given by right hand thumb rule or right handed screw rule as shown in the figure:



35.

Ans: To demagnetise a permanent magnet, it is placed inside a long solenoid whose axis is aligned east–west, as shown in the diagram. A 12 V AC supply is connected in series with a rheostat.



When AC is switched on, the direction of current keeps changing rapidly. In one second, the polarity reverses many times, causing the magnetic domains of the magnet to continuously reverse their alignment. These repeated reversals prevent the domains from settling in any fixed direction. As a result, they become randomly oriented and the magnet gradually loses its magnetism.

Finally, the current is slowly reduced to zero using the rheostat, making it even harder for the domains to realign. Thus, the magnet becomes completely demagnetised.

36.

Ans:

(a) Given, $i = 0.3 \text{ A}$, $V = 6.0 \text{ Volt}$, $R_{\text{total}} = ?$

$$\begin{aligned} R_{\text{total}} &= \frac{V}{i} \\ &= \frac{6.0 \text{ V}}{0.3 \text{ A}} = 20 \Omega \end{aligned}$$

(b) Resistance R and 60Ω are in parallel combination

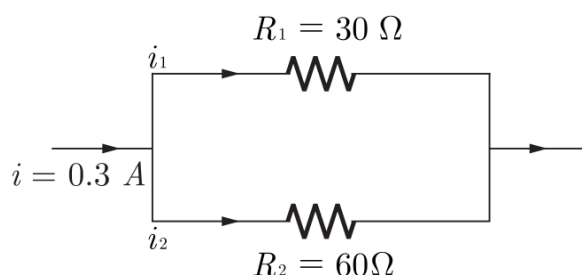
Therefore, $20 \Omega = \frac{R \times 60}{R + 60}$

$$20 R + 1200 = 60 R$$

$$40 R = 1200$$

$$R = \frac{1200}{40} = 30 \Omega$$

(c) Let the current $i = 0.3 \text{ A}$ be distributed in the two resistors $R_1 = 30 \Omega$ and $R_2 = 60 \Omega$ as i_1 and i_2 respectively.



i.e. $i = i_1 + i_2 \quad \dots(1)$

The current,

$$i_1 = \left(\frac{R_2}{R_1 + R_2} \right) i$$

$$= \left(\frac{60}{30 + 60} \right) \times 0.3$$

$$= \frac{60}{90} \times 0.3$$

$$i_1 = 0.2 \text{ A}$$

From equation (1), the current i_2 in 60Ω resistor

$$= i - i_1$$

$$= 0.3 \text{ A} - 0.2 \text{ A} = 0.1 \text{ A}$$

Note: Here i_2 may also be calculated using

$$i_2 = \left(\frac{R_1}{R_1 + R_2} \right) i$$

37.

Ans: (c) a straight line passing through origin

Explanation:

For an **ohmic resistor**, the potential difference (V) is directly proportional to the current (I), i.e.,

$$V \propto I$$

This means the graph of V against I is a **straight line** and it must **pass through the origin**, because when current is zero, voltage is also zero.

Hence, the correct graph is **a straight line passing through the origin**.

38.

Ans: (c) a live as well as neutral wire

Explanation:

In the main distribution board, the **main switch must control both the live wire and the neutral wire.**

This ensures that when the switch is off, the appliance or circuit is completely disconnected from the supply for safety.

So, the correct option is **(c)**.

39.

Ans: (d) a current flows in the circuit

Explanation:

A coil wound around soft iron becomes an **electromagnet only when electric current passes through it.**

The flowing current creates a magnetic field, which magnetises the soft iron core.

So, the correct option is **(d)**.

40.

Ans: (b) the needle will deflect towards East

Explanation:

The wire ***AB*** is placed in the **North–South** direction, and the magnetic needle is placed **just below the wire.**

When the key ***K*** is closed, current flows through the wire.

Using the **Right-hand thumb rule**, the magnetic field produced around the conductor will act **from West to East** at the position of the needle.

Because of this magnetic field, the magnetic needle experiences a force and **deflects towards the East.**

So, the correct option is **(b)**.

41.

Ans:

(i) The switch **S₁** is **open**, because no current is flowing through the circuit.

(ii) Since no wire from the switch is connected to complete the circuit, no current flows. Therefore, the circuit is incomplete, which means the **switch must be open**.

(iii) The magnetic compass is placed in the setup to **observe the deflection** and thus determine the **presence and direction of current** flowing through the coil.

42.

Ans:

(i) When the wire is stretched to three times its original length, its **volume remains constant**.

$$V = Al = A'l'$$

Since $l' = 3l$:

$$A' = \frac{Al}{3l} = \frac{A}{3}$$

The original resistance is:

$$R = \frac{\rho l}{A}$$

The new resistance is:

$$R' = \frac{\rho l'}{A'} = \frac{\rho(3l)}{A/3} = \frac{3\rho l}{A/3} = 9 \left(\frac{\rho l}{A} \right) = 9R$$

Given $R = 2\Omega$:

$$R' = 9 \times 2 = 18\Omega$$

(ii) The **specific resistance (resistivity)** depends only on the nature of the material and does not depend on the dimensions of the wire.

Therefore, the **specific resistance remains unchanged**.

43.

Ans: Heat dissipated in a conductor depends on:

- (i) the square of the current flowing through the conductor,
- (ii) the resistance of the conductor, and
- (iii) the time for which the current flows.

Thus,

$$Q = I^2 R t$$

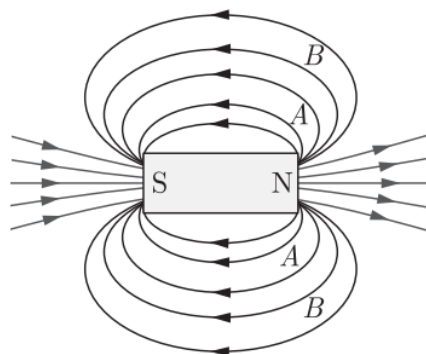
44.

Ans:

- (i) The pole pieces are made cylindrical so that the plane of the coil always remains in the direction of the magnetic field lines.
- (ii) A soft iron core is placed inside the coil because it increases (intensifies) the magnetic field and makes it radial by concentrating the field lines, due to its high permeability.

45.

Ans: A solenoid is a cylindrical coil consisting of many tightly wound turns of insulated wire, with its length greater than its diameter.



When an electric current is passed through a solenoid, a magnetic field is produced. Since the current in each circular turn flows in the same direction, the magnetic fields of all the turns combine. As a result, one end of the solenoid behaves like the **North pole** of a bar magnet and the other end behaves like the **South pole**.

Thus, a solenoid acts like a bar magnet, as shown in the figure.

46.

Ans:

Given, $I = 4 \text{ A}$, $V = 12 \text{ Volt}$

and $t = 10 \text{ min} = 600 \text{ s}$

$$\begin{aligned}\text{Total energy consumed} &= V \times I \times t \\ &= 12 \times 4 \times 600 \\ &= 28800 \text{ J} = 28.8 \text{ KJ}\end{aligned}$$

47.

Ans: MCB stands for **Miniature Circuit Breaker**. It is a device that automatically switches off the mains supply when the current exceeds the safe limit. This prevents damage to the circuit and protects appliances from short circuits and overloading.

The MCB replaces the **main fuse** in a domestic electric circuit. It is more reliable, easier to operate, and more convenient than a fuse.

48.

Ans:

(i) The resistivity of a **metallic wire increases** when the temperature increases.

(ii) The resistivity of a **semiconductor decreases** as the temperature increases.

(iii) The resistivity of an **alloy (e.g., constantan)** remains almost unchanged with variation in temperature.

49.

Ans:

Given:

$$R_1 = 3 \Omega, \quad R_2 = 6 \Omega$$

For resistors in parallel:

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2}$$

So,

$$\frac{1}{R_P} = \frac{1}{3} + \frac{1}{6} = \frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

Therefore,

$$R_P = 2 \, \Omega$$

50.

Ans:

Given:

$$E = 6 \, \text{V}, \quad i = 0.5 \, \text{A}$$

Time:

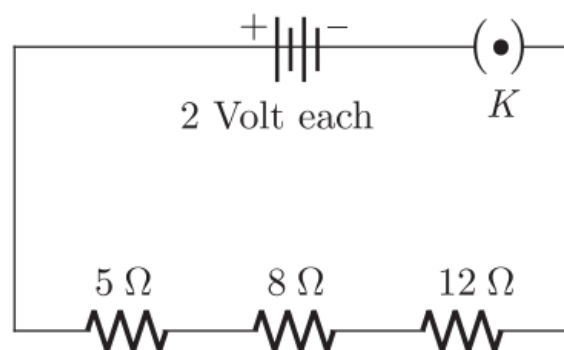
$$t = 5 \, \text{min} = 300 \, \text{s}$$

Energy supplied by the battery:

$$\begin{aligned} \text{Energy} &= E \times i \times t \\ &= 6 \, \text{V} \times 0.5 \, \text{A} \times 300 \, \text{s} \\ &= 900 \, \text{J} \end{aligned}$$

51.

Ans:

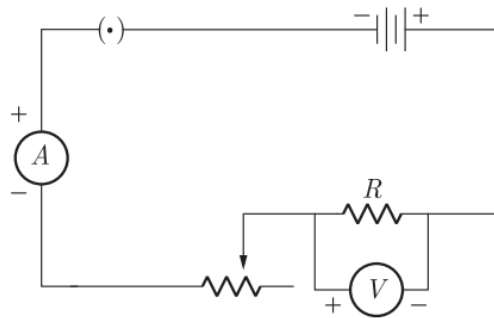


52.

Ans: Ohm's law states that, **if the physical conditions of a conductor remain unchanged**, the electric current flowing through it is **directly proportional** to the potential difference across its ends.

53.

Ans:



54.

Ans:

From the figure, resistors **R₁, R₂ and R₆** are connected in series.

So, their combined resistance is:

$$R = R_1 + R_2 + R_6 = 1 + 1 + 1 = 3\Omega$$

Similarly, **R₄, R₅ and R₃** are in series. Therefore,

$$R' = R_4 + R_5 + R_3 = 1 + 1 + 1 = 3\Omega$$

These two combinations (**R** and **R'**) are connected **in parallel**.

Hence, equivalent resistance:

$$\frac{1}{R_P} = \frac{1}{R} + \frac{1}{R'} = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

$$R_P = \frac{3}{2} = 1.5 \Omega$$

55.

Ans:

Given:

Power of motor (1 hp) = **746 W**

Power of one bulb:

$$P = V \times I = 220 \times 0.2 = 44 \text{ W}$$

Power of 5 bulbs:

$$5 \times 44 = 220 \text{ W}$$

Power of two fans:

$$2 \times 80 = 160 \text{ W}$$

Total power of all appliances:

$$746 + 220 + 160 = 1126 \text{ W}$$

All appliances work for 5 hours daily.

Energy consumed per day:

$$1126 \times 5 = 5630 \text{ Wh}$$

Energy consumed in one month (30 days):

$$30 \times 5630 = 168900 \text{ Wh}$$

Convert to kWh (units):

$$\frac{168900}{1000} = 168.9 \text{ kWh}$$

Cost of energy at 20 paise per unit:

$$\text{Cost} = 0.20 \times 168.9 = ₹33.78$$

56.

Ans: When an electric current flows through a conductor, the conductor offers resistance to the motion of electrons. Because of this resistance, some electrical energy is converted into heat energy. To find this heat produced, we calculate the work done in sending current through the conductor.

If a potential difference of **V volts** is applied across the conductor and a current **i** flows for **t seconds**, the charge flowing is:

$$Q = it$$

By definition of potential, the work done in moving charge Q through potential difference V is:

$$W = QV = itV \quad \dots(1)$$

Using Ohm's law, $V = iR$:

$$W = (iR) it = i^2 Rt \quad \dots(2)$$

Also, since $i = \frac{V}{R}$:

$$W = \left(\frac{V}{R}\right)^2 Rt = \frac{V^2}{R}t \quad \dots(3)$$

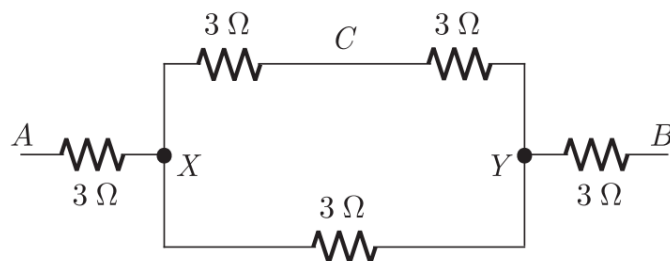
Thus, the electrical energy produced in the circuit is:

$$W = Vit = i^2 Rt = \frac{V^2}{R}t$$

These expressions give the electrical energy supplied by the cell or mains when current i flows for time t .

57.

Ans: The equivalent circuit for the above circuit is as follows :



(i) The total resistance between X and Y is given by

$$\begin{aligned} \frac{1}{R} &= \frac{1}{6} + \frac{1}{3} \\ &= \frac{1+2}{6} = \frac{3}{6} = \frac{1}{2} \Omega \end{aligned}$$

Hence, $R = 2 \Omega$

(ii) Total resistance between A and B is given by

$$R' = 3 + 2 + 3 = 8 \Omega$$

58.

Ans: (d) all of these

Explanation:

The power of a d.c. motor depends on the **torque** it produces. Torque can be increased in the following ways:

- **Increasing the current** in the coil increases the magnetic force and hence torque.
- **Increasing the number of turns** of the coil increases the magnetic field, which also increases torque.
- **Using a laminated soft iron core** strengthens the magnetic field inside the motor and improves its efficiency.

Since **all** these factors increase the power of the motor, the correct answer is **(d)**.

59.

Ans: (a) mechanical energy changes to heat energy.

Explanation:

An electric motor **converts electrical energy into mechanical energy**.

The electrical energy supplied to the motor is used to produce rotational motion (rotational kinetic energy), which is a form of mechanical energy.

Thus, **(a) is the correct option**.

60.

Ans: The potential difference between two points is said to be **1 volt** when **1 joule of work** is done to move **1 coulomb of charge** from one point to the other.

61.

Ans: Electrostatic potential at a point is the amount of work done in bringing a unit positive charge from infinity to that point.

62.

Ans: Connect the two ends of the heating coil together and apply the 110 V supply between this joined end and the midpoint of the coil.

This divides the coil into two equal halves, each having resistance **$R/2$** (half of the original resistance).

Each half now consumes power

$$P' = \frac{P}{2}$$

i.e., half of the original power.

Since the two halves are connected **in parallel**, the total power consumed becomes

$$P = P' + P' = P$$

which is the same as before.

Thus, the heater can operate on 110 V **without changing coil length and without change in power consumption.**

63.

Ans: Differences between Electromotive Force (e.m.f.) and Potential Difference (p.d.):

Electromotive Force (e.m.f.)	Potential Difference (p.d.)
(i) It is the maximum potential difference between the terminals of a cell when no current is drawn (open circuit).	(i) It is the potential difference between any two points of a conductor when current flows (closed circuit).
(ii) It is the source of electrical energy in a circuit.	(ii) It is measured between any two points of an electric circuit.
(iii) It depends on the electrodes and the electrolyte of the cell.	(iii) It depends on the resistance between the two points in the circuit and the current flowing.
(iv) It is always greater than the potential difference across any circuit element.	(iv) It is always less than the e.m.f. of the cell.

64.

Ans: (a) Potential is defined as the amount of work done in bringing a single positive charge from infinity to that point. Its unit is volt.

(b) The electromotive force of a cell is equal to its chemical energy that is converted into electrical energy (or light, heat, etc.). It can be defined as the maximum potential difference across the terminals of a cell or open circuit potential difference since in that condition no energy will be lost within it. The unit of emf is volt and it is a measure of the work done by the cell per unit charge.

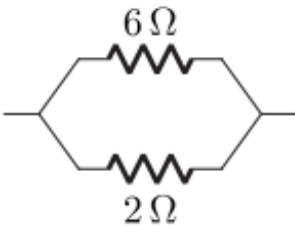
(c) Potential difference across a conductor is defined as the amount of work done in moving one coulomb charge from one end to the other.

$$\text{Potential difference, } (V) = \frac{\text{Work Done } (W)}{\text{Charge } (q)}$$

Its unit is also volt.

65.

Ans:

	Series combinations	Parallel combinations
(i)		
(ii)	In this combination, potential difference across the combination equals the sum of the potential differences across each of the resistors.	In this combination, potential difference across the combination is equal to the potential difference across any one.
(iii)	The current passing through each resistor has the same value.	The total current in the battery circuit is the sum of the currents in each resistor.

66.

Ans:

(i) Total current consumed

Total power of the lamps:

$$P = 200\text{ W} + 100\text{ W} + 60\text{ W} + 40\text{ W} = 400\text{ W}$$

Current:

$$I = \frac{P}{V} = \frac{400}{220} = \frac{20}{11} = 1.8\text{ A}$$

(ii) Total resistance of the arrangement

$$R = \frac{V}{I} = \frac{220}{20/11} = 121\Omega$$

(iii) Cost of keeping them lighted for 7 hours daily for 30 days

Energy consumed:

$$\text{Energy} = \frac{400 \times 7 \times 30}{1000} = 84\text{ kWh}$$

Total cost (at 30 paise per unit):

$$\text{Cost} = 84 \times \frac{30}{100} = ₹25.20$$

67.

Ans:

The potential difference across the $5\ \Omega$ resistor = **10 V**

The potential difference across the $3\ \Omega$ resistor = **6 V**

Given resistor $R = 5\ \Omega$

(i) Current through the $5\ \Omega$ resistor

$$I = \frac{V}{R} = \frac{10}{5} = 2\text{ A}$$

(ii) Current through R

Since both resistors are in **series**,

current is the same through each resistor.

$$I = 2 \text{ A}$$

(iii) Value of R

$$R = \frac{V}{I} = \frac{6}{2} = 3 \Omega$$

(iv) Value of V

Total resistance in series:

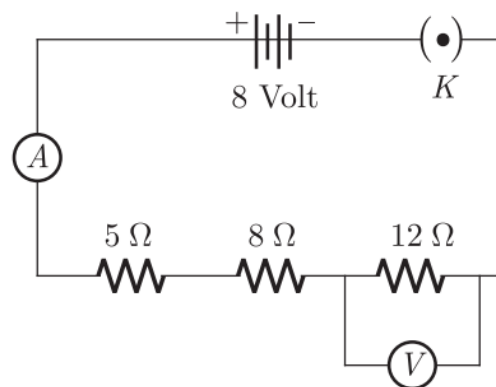
$$R_{\text{total}} = 5 + 3 = 8 \Omega$$

Using $V = IR$:

$$V = 2 \times 8 = 16 \text{ V}$$

68.

Ans:



Since the resistors are connected in series, therefore, the equivalent resistance of the combination is

$$R_s = R_1 + R_2 + R_3 = 5 + 8 + 12 = 25 \Omega$$

Now, current in the circuit

$$I = \frac{V}{R_s} = \frac{8}{25} = 0.32 \text{ A}$$

Voltage across the 12 ohm resistor

$$V = IR = 0.32 \times 12 = 3.84 \text{ V}$$

69.

Ans:

Given:

Heat capacity of calorimeter, $c = 50 \text{ J}^\circ \text{C}^{-1}$

Mass of liquid, $M = 1.0 \text{ kg}$

Specific heat capacity of liquid, $C = 450 \text{ J kg}^{-1}^\circ \text{C}^{-1}$

Rise in temperature, $(T_2 - T_1) = 10^\circ \text{C}$

Current, $I = 2.0 \text{ A}$

Time, $t = 10 \text{ min} = 600 \text{ s}$

(i) Resistance of the coil

Total heat produced = Heat absorbed by calorimeter + Heat absorbed by liquid

$$\begin{aligned}\text{Heat} &= (mc + MC)(T_2 - T_1) \\ &= (50 + 1.0 \times 450) \times 10 \\ &= (50 + 450) \times 10 = 5000 \text{ J}\end{aligned}$$

Electrical heat produced

$$\begin{aligned}H &= I^2 R t \\ 5000 &= (2)^2 R \times 600 \\ 5000 &= 2400 R \\ R &= \frac{5000}{2400} \approx 2.08 \Omega\end{aligned}$$

So, resistance of the coil = **2.08 Ω** .

(ii) Potential difference across the coil

$$V = IR = 2 \times 2.08 = 4.16 \text{ V}$$

So, potential difference = **4.16 V**.

Assumption used:

There is **no heat loss to the surroundings**.